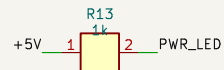
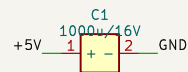
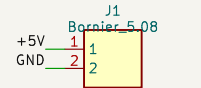
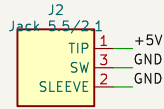
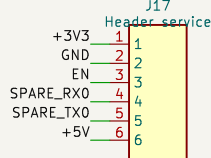


A PROTEGER EN AMONT (fusible/disjoncteur) : charges inductives : snubber cote charge.

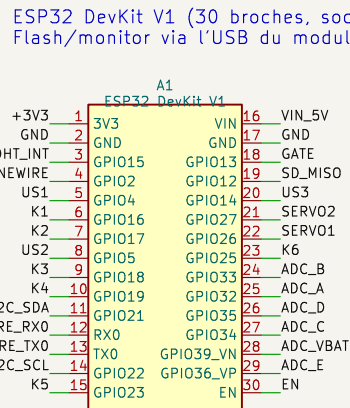
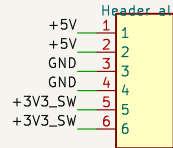
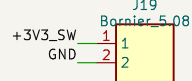
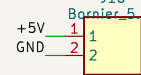
ALIMENTATION 5V (3A recommandé) – jack OU bornier.
D5 protège le rail si l'USB du DevKit est branché en même temps.



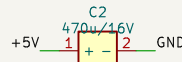
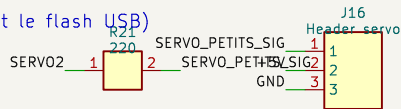
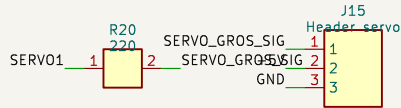
TOUS les GPIO libres sur header (Dupont),
y compris RX0/TX0 (les laisser libres pendant le flash USB)
et GPIO36/39 (entrees seules).



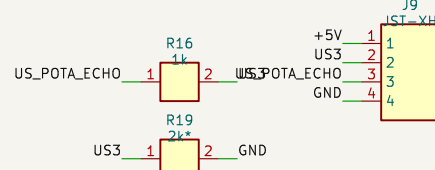
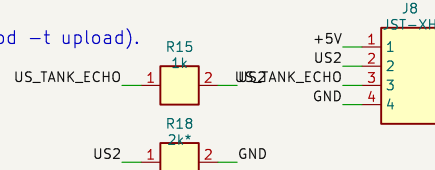
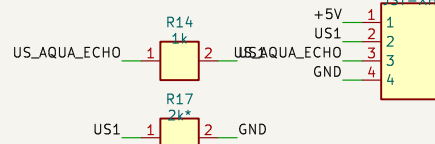
Distribution 5V / 3V3—capteurs / GND :
borniers à vis + rail header.



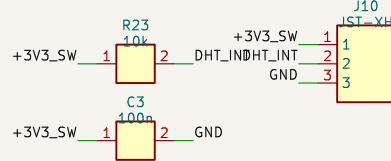
SERVO NOURRISSEURS 5V
GPIO12 (strapping MTDI) : pull-down R22, signal série R20.



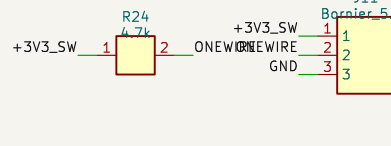
3x HC-SR04 (5V), mode mono-broche TRIG=1 ECHO=2 (sensor_ultrasonic.cpp).
Echo 5V ramené à 3V3 par pont 1k/2k sur chaque canal.



AIR : DHT11 (option DHT22, -DUSE_DHT22) - pull-up 10k.



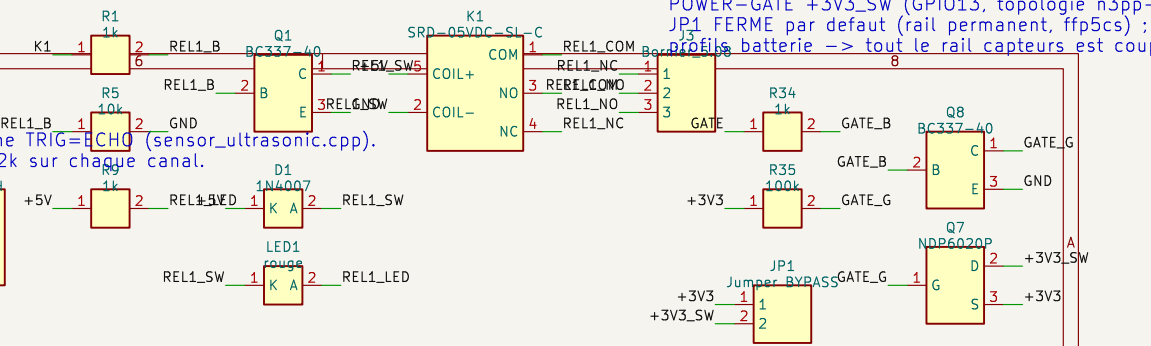
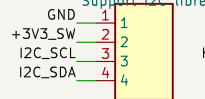
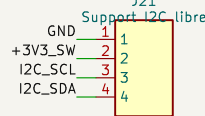
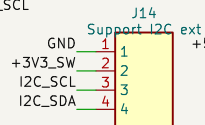
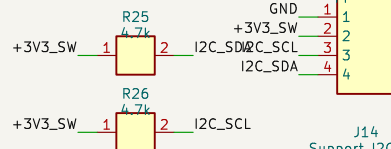
EAU : DS18B20 (1-Wire, pull-up 4.7k).



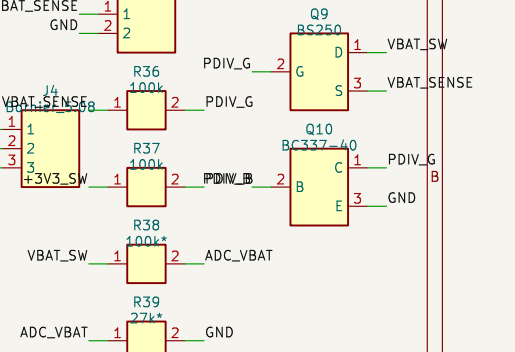
LUMIERE : LDR déportée + 10k (GPIO34 = entrée seule).



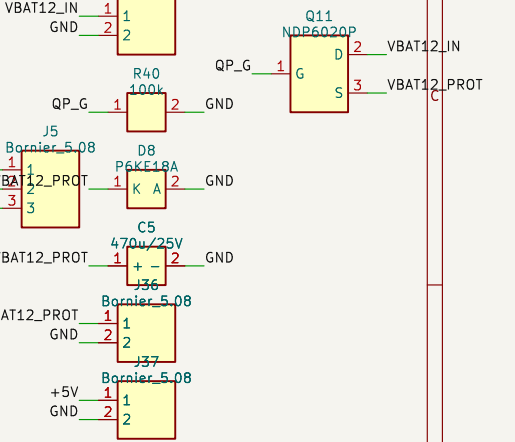
I2C : OLED SSD1306 0x3C + header extension (DS3231 si besoin)



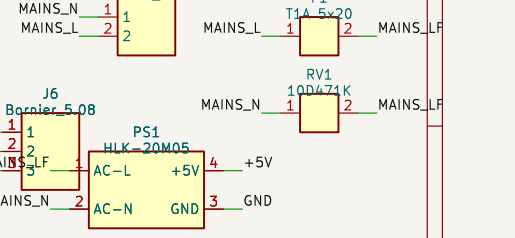
PONT DIVISEUR VBAT COMMUTE : actif seulement
R38/R39 SEON PROFIL : 1S Li-ion = 100k/100



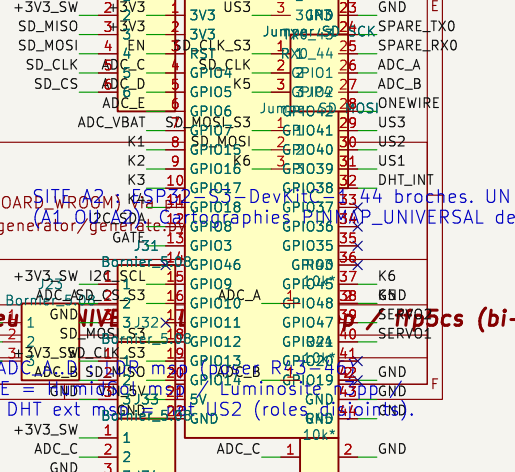
PROFIL BUS 12V SOLAIRE : fusible lame 7,5-10A
anti-inversion P-MOSFET, TVS 18V, réservoir ; b
(module MP1584/XL4015 faible Iq sur entretoise



PROFIL SECTEUR : Hi-Link HLK-20M05 EMBARQUÉ
+ varistance 300VAC. Le corps du module enjambe
secteur/logique (fente fraisée dessous). ZONE 2.



microSD : module SPI 3V3 sockete. JP2/3/4 : s
CS/SCK/MOSI = 53 stat (1-2, default) ou WR0
MISO = 1 het partage CS (A1) - GPIO12 / A2-IO

[illegible]